

Rule-Based or LLM Agent? Real-Time Behaviour Selection for an Emotion-Responsive Reachy Mini

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Abstract—Outline (to be written last, 150–200 words):

- **Social robots must react promptly and appropriately to human emotion; LLM agents promise flexible behaviour but are slow and non-deterministic.**
- **Research question: can a local LLM agent select behaviour on the Reachy Mini in real time, and what does it offer over rules?**
- **Method: fixed perception (MediaPipe), action repertoire and motion layer; only the behaviour-selection layer is exchanged (rules, LangChain agent with tool calling, structured output, hybrid); recorded and annotated stimulus, ten runs each.**
- **Results: latency, resource demand, validity, consistency and appropriateness per configuration (after measurement).**
- **Conclusion: answer to the research question and recommendation.**

I. INTRODUCTION

Robots are moving from industrial cells into everyday environments such as homes, classrooms and offices. In these settings the quality of an interaction depends less on task performance than on whether the robot is perceived as responsive and approachable. People respond empathically even to robots [20], and observers rate a robot as simple as the Reachy Mini as warmer and more sociable when it expresses positive affect through head and antenna motion alone [18]. Whether such a reaction is read as a response to one’s own behaviour, however, depends on its timing [19]. For a social robot, prompt and appropriate emotional responsiveness is therefore a core function rather than an add-on.

Problem statement. The component that turns perceived human state into robot behaviour – the *behaviour-selection layer* – is traditionally implemented with hand-written rules, state machines or behaviour trees [5]. Such controllers are fast, predictable and easy to verify, but every situation has to be anticipated and encoded by hand, and repeated situations lead to identical, mechanical reactions. Large language models (LLMs) used as tool-calling agents [22] offer an alternative: the robot’s reaction is selected from a natural-language description of the situation and the interaction history, which promises more varied and context-sensitive behaviour without programming every case. Frameworks such as LangChain [10] and local inference servers such as Ollama [15] make this approach available on commodity hardware and without a cloud connection. However, LLM inference takes hundreds

of milliseconds to seconds, occupies substantial compute and memory, and does not guarantee valid or reproducible outputs. These properties collide with the requirement of contingency: users prefer robot responses within about one second [19], and a reaction that arrives too late is no longer read as a response to one’s own behaviour. In robotics, LLMs have so far mainly been used for task planning without hard timing constraints [1], [11]. For reactive, emotion-driven behaviour on resource-constrained hardware it is unclear whether an LLM agent is a viable replacement for rule-based control, and a controlled comparison that considers timing, resource cost and behavioural quality together is missing.

Research question. This paper therefore addresses the question:

Can a local LLM agent select emotion-responsive behaviour on the Reachy Mini in real time, and what does it offer over a rule-based controller?

The first part is answered by the latency of the agent relative to the contingency threshold, the second by its resource demand and by the validity, consistency and appropriateness of the chosen reactions. We expect (H1) the rule-based controller to be faster by at least one order of magnitude, with the agent exceeding the threshold in a relevant share of reactions; (H2) the agent to require considerably more CPU/GPU time and memory; (H3) the agent to produce more varied reactions at the cost of lower consistency and occasional invalid actions; and (H4) a hybrid, in which rules handle time-critical reflexes and the agent handles the remaining events, to retain the latency of rules for those reflexes.

Contribution. We implement a behaviour architecture for the Reachy Mini [17] in which perception, action repertoire and motion execution are identical across conditions and only the behaviour-selection layer is exchanged. On this basis we compare four configurations – a rule-based controller, a LangChain agent with tool calling, a single LLM call with structured output, and a hybrid – on an identical, recorded and annotated stimulus. The contributions are (i) an open, reproducible test bed with instrumented timestamps from camera frame to motor command, (ii) a quantitative comparison of latency, resource demand and behavioural quality, and (iii) design recommendations for when an LLM agent is worth its cost in reactive social robotics.

Section II reviews the background, Section III describes architecture, configurations and evaluation design, Section IV reports the measurements, and Section V concludes.

II. STATE OF THE ART

A. Emotional Responsiveness in Social Robots

- Humans empathise with robots; EEG shows pain-related responses to robot hands [20] → emotional reactions of a robot are socially meaningful.
- Response timing: user preference peaks at about 1 s response time; fillers mitigate longer delays [19] → requirement: contingency; threshold for evaluation.
- Reactions must fit the displayed emotion → requirement: appropriateness.
- Design: baby-schema features increase perceived cuteness and caretaking motivation [7]; moderate rather than near-human likeness avoids the uncanny valley [14] → Reachy Mini's stylised morphology.
- Reachy Mini: positive expressions via head and antennas alone rated warmer and more sociable than negative ones [18] → reduced expressive channel is sufficient.
- Repeated identical reactions appear mechanical → requirement: variability.

B. Rule-Based Behaviour Control

- Finite state machines, behaviour trees [5], lookup tables as standard in social robots.
- Strengths: deterministic, fast (μ s), verifiable, no extra hardware.
- Weaknesses: every case must be modelled; rigid; repetitive; little use of interaction history.

C. LLM Agents for Robot Behaviour

- LLMs as high-level planners: SayCan [1], Code as Policies [11] – no hard real-time constraints.
- Agent pattern: reasoning + tool calls [22]; frameworks such as LangChain [10].
- Small open models with tool calling (e.g. Qwen2.5 [21]) run locally via Ollama [15].
- Known issues: latency, invalid or hallucinated actions, non-reproducibility, resource demand.
- LLMs generate expressive nonverbal robot behaviour from few-shot prompts [13]; real-time reactivity and cost not the focus.

D. Real-Time Perception Toolkits

- Pre-trained, CPU-capable pipelines instead of task-specific training.
- MediaPipe [12]: face mesh with blendshapes, hand gestures.
- Vosk [2]: offline speech recognition.
- Held constant in all configurations of this work.

E. Research Gap

- LLM agents in robotics evaluated for planning, not for contingent emotional reactions.
- No controlled comparison of rules vs. LLM agent that measures latency, resources and behaviour quality together on constrained hardware.
- ⇒ this work: identical pipeline, only behaviour selection exchanged.

III. MATERIALS AND METHODS

A. Platform

- Reachy Mini: 6-DoF head (Stewart platform), body yaw, two antennas, camera, microphones; no facial actuators.
- Python SDK + daemon; physical robot or MuJoCo simulation [8].
- Host computer for perception and LLM: **[TODO: CPU/GPU/RAM aneben]**.

B. System Architecture

- Three layers (Fig. 1): perception → behaviour selection → motion execution.
- Perception (camera rate): FaceLandmarker blendshapes → four affective categories; rectangle of eye and mouth corners → gaze target and distance class; gesture recogniser + custom wave detector; optional speech.
- Event detector: only relevant changes are passed on (person appears/leaves, stable emotion or distance change, gesture, utterance); debounce 0.5 s.
- Behaviour selection in a separate worker thread; events queued (FIFO).
- Motion execution at 50 Hz: head follows face in all configurations; expressions overlaid as keyframes with minimum-jerk interpolation [6]; non-blocking.
- Consequence: tracking stays real-time even while the agent is computing.



Fig. 1: Placeholder. Three-layer architecture; only the behaviour-selection layer differs between configurations.

C. Action Repertoire

- Shared by all configurations (Tab. 3): seven emotional postures (neutral, joy, fear, curious, sad, cautious, confused), two gestures (wave, nod), *no action*.
- Action = antenna angles + head offsets + duration; several pose variants per emotion, chosen at random.
- Same list serves as rule targets and as agent tools/schema values.

D. Behaviour-Selection Configurations

- **R – rules:** lookup table (event, emotion, distance) → action; mirror joy, fear at anger or intrusion, curiosity at surprise, return waves.
- **A-TC – agent with tool calling:** LangChain `create_agent`, one tool per action type, textual observation + last five events as history, persona prompt, tool ends the agent loop directly.
- **A-SO – structured output:** same prompt and model, one call returning a schema-validated action object; isolates the cost of the agent loop.
- **H – hybrid:** rules for time-critical events (appear, leave, gesture, distance), A-SO for emotion changes and speech.
- **Model:** local via Ollama, default Qwen2.5-3B [21], temperature 0; optional comparison of model sizes.

Cfg	Selection	Isolated effect
R	rules	baseline
A-TC	agent, tool call	rules → agent
A-SO	LLM, schema	cost of the agent loop (vs. A-TC)
H	rules + LLM	can a hybrid combine both?

Tab. 1: Compared behaviour-selection configurations. Perception, action repertoire and motion execution are identical in all of them.

E. Requirements and Metrics

- Mapping of requirements from Section II to metrics in Tab. 2.
- Timestamps: t_0 frame read, t_1 selection started, t_2 first action, t_3 first motor command.

Requirement	Metric	Reported in
Contingency	Decision latency $t_2 - t_1$, end-to-end latency $t_3 - t_0$ (median, IQR, p95); share above 1 s	Sec. IV-A
Efficiency	CPU, GPU, RAM of process and LLM server	Sec. IV-B
Appropriateness	Agreement with annotated acceptable reactions; rate of invalid or missing actions	Sec. IV-C
Variability vs. consistency	Share of identical decisions across runs; distinct actions per event type	Sec. IV-D

Tab. 2: Requirements, metrics and where they are reported.

F. Evaluation Design

- One recorded stimulus video (person enters, smiles, waves, approaches, looks angry, speaks, leaves), replayed in real time into every configuration.
- Annotation: event time windows and set of acceptable reactions; **[TODO: Anzahl Annotatoren, Uebereinstimmung]**.
- Ten runs per configuration; warm-up call discarded.

- Resource sampling every 0.5 s in parallel.
- Evaluation script produces tables and plots per configuration.

G. Software Environment

- Python 3.12, MediaPipe, OpenCV [16], NumPy [9], LangChain, Ollama, Reachy Mini SDK.
- Code and evaluation scripts in the project repository **[TODO: Link, Versionen]**.

IV. EXPERIMENTAL RESULTS

Planned content – measurements pending.

A. Latency

- Box plot of decision and end-to-end latency per configuration (Fig. 2), 1 s threshold marked.
- Share of reactions above threshold; queueing delay when events arrive during an ongoing decision.
- A-TC vs. A-SO: overhead of the agent loop.



Fig. 2: Placeholder. Decision and end-to-end latency per configuration.

B. Resource Demand

- Mean and peak CPU, GPU and RAM per configuration; share caused by the LLM server.
- Does the agent fit on the target hardware next to perception?

C. Behavioural Appropriateness

- Share of appropriate reactions per configuration and event type.
- Invalid or missing actions of the agent variants.
- One or two qualitative examples of context-sensitive (or wrong) agent decisions.

D. Consistency and Variability

- Share of identical decisions over ten runs; distinct actions per event type.
- Is the agent's variability useful (variety) or harmful (contradictory)?

E. Discussion

- Answer to both parts of the research question; hypotheses H1–H4 confirmed or rejected.
- Trade-off latency/resources vs. flexibility; role of the hybrid.
- Limitations: one stimulus, one hardware setup, no user study.

V. SUMMARY AND OUTLOOK

- Summary: comparison of rules and LLM agent as behaviour selection on the Reachy Mini; key numbers; answer to the research question.
- Outlook: user study with Godspeed [3] or RoSAS [4]; streaming and speculative execution to reduce latency; memory across interactions; vision-language model instead of symbolic state.

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APPENDIX

Action	Antennas [°]	Head offset	Dur. [s]
neutral	−30/30	–	1.0
joy	−90/90	pitch −10, z +10	0.6
fear	−180/180	pitch +10, z −30	1.0
curious	−120/110	yaw ± 20 , roll ± 10	1.2
sad	−10/10	pitch +20, z −15	1.5
cautious	−120/110	–	1.0
confused	−90/30	yaw ± 20 , roll ± 15	0.5
wave	$0 \leftrightarrow \mp 90$ ($2\times$)	–	5×0.25
nod	–	pitch 15/−5/15/0	4×0.2

Tab. 3: Action repertoire shared by all configurations (from reachy_hri/actions.py).